
api-gateway (:4000)

Request flow, file by file

150 questions with answers

12 sections

Event-driven microservices · Go · Kafka · Redis

3 — api-gateway (:4000)

Job: one front door. Verify the token once, rate limit, route to one of eight services.

```
browser
| POST /api/donations    Authorization: Bearer <jwt>
▼
index.js                ▶ starts the server
▼
app.js
|
|— helmet() · cors()
|
| (!) NO express.json() – deliberately.
|     Parsing the body would CONSUME the stream and the proxy
|     would forward an empty body downstream.
|
|— middleware/requestContext.js    ▶ x-trace-id
|
|— config/routes.js                ▶ match prefix, IN ORDER
|   /api/auth      → auth-service    basePath /auth
|   /api/donations → donation-service basePath /donations
|   /uploads       → donation-service basePath /uploads
|   /api/geo       → geocoding-service basePath ''
|   /api/location  → agent-location   basePath ''
|   /api/engine    → assignment-engine basePath ''
|   /api/monitoring → tracking-service basePath /monitoring
|   /api/tracking  → tracking-service basePath /tracking
|   /api/notify    → notification     basePath /notifications
|   /api/analytics → analytics        (optional)
|
| (!) /api/monitoring MUST be registered before /api/tracking,
|     or /api/tracking swallows every monitoring request.
|
|— ratelimit                300/min general
|                           20/min on /api/auth, failures only
|
|— middleware/verifyToken.js
|   ▶ jwt.verify(token, secret, { algorithms: ['HS256'] })
|   ▶ sets req.user          ▶ auth:false routes skip this
|
|— createProxyMiddleware
|   ▶ pathRewrite: basePath + path    ← NOT a regex substitution
|   ▶ forwards x-trace-id, x-user-id, x-user-role
|   ▶ on error → 502 with an explanation
|   |
|   ▼
|   donation-service :4002 POST /donations
```

The bug worth remembering. `app.use(prefix, ...)` **strips** the prefix before the middleware runs, so `/api/auth/signup` was already `/signup` by the time the proxy saw it — and a `pathRewrite` regex matching `^/api/auth` matched nothing. 16 of 28 checks failed. The fix is to *prepend* the service's own `basePath` rather than substitute the public prefix.

GET /ready does not proxy: it fans out to all seven services and returns 503 if any is down.